Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/21/2007
PI Name:	Cartreine, James A. Ph.D.		
Project Title:	Self-guided Depression Treatment on Long-duration Space Flights: A Continuation Study		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jcartreine@partners.org	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-851-8913
Organization Name:	Brigham and Women's Hospital/Harvard Medical Center		
PI Address 1:	Program in Behavioral Informatics and eHealth		
PI Address 2:	Department of Psychiatry		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:	NOTE: PI moved to Brigham and Women's Hospital; formerly at Beth Israel Deaconess Medical Center, Boston, MA, per info received December 2011 (Ed.) Name change to Cartreine in summer 2008 (from Carter), per NSBRI information (11/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	01/01/2005	**End Date:**	08/31/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Hegel, Mark (Dartmouth College) Slack, Warner (Harvard University) Locke, Steven (Harvard Medical School)		
Grant/Contract No.:	NCC 9-58-NBPF00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Objective: Depression can present a significant threat to long-duration space missions and crews need the ability to recognize it and treat it effectively. In our current NSBRI study, Designing a Smart Medical System for Psychosocial Support on Long-duration Space flights, we are developing a prototype computer-based system that includes a module for the recognition and treatment of depression. Before being used by astronauts, this system needs to be further developed and tested for efficacy in an analog population. The countermeasure we are developing is intended for use both in pre-flight training and on orbit on the ISS, and has substantial Earth benefits. Specific aims of this proposal are: 1. Complete development of the module for recognizing and preventing depression. We will complete this module, moving from the prototype stage to a completed computer-based tool. We will modify the design based on data from our usability and acceptability study with astronauts at Johnson Space Center. We will pilot-test the module with 5 participants, and make revisions based on their input. 2. Develop the module for implementing computer-based Problem-Solving Treatment. Currently, problem-solving treatment is done using a practitioner who guides the patient through this structured therapy and follows their progress. These practitioners are trained to administer the therapy through a formalized training program that incorporates a workbook and instructional videos1. These materials are well suited for presentation via multimedia on computer to make the treatment self-guided. The PST intervention will be modified from its current video/workbook form to a computer-based form that can be used by astronauts or other professionals who are suffering from depression. 3. To evaluate the efficacy of the computer-delivered Problem-Solving Treatment (PST) module for depression in a randomized controlled trial. Hypotheses related to this aim are that, compared to a waiting-list control: a. Individuals receiving computer-delivered PST will show a greater reduction in depression symptoms, as rated by a clinician. b. Individuals receiving computer-delivered PST will show a greater reduction in depression, as rated by self-report. c. A greater proportion of individuals receiving computer-delivered PST will have remitted at post-treatment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Depression is widespread, and effective treatments are not available to all persons who have it. This computer-based depression treatment, with simple modifications and revisions, could be adapted for use in other isolated operational environments, such as polar research stations, submarines, commercial ships, oilrigs, and underwater research bases. Furthermore, even greater value could be derived by making similar psychosocial support systems available to the public in settings such as primary care practices, public and mental health centers, schools, social services offices, places of worship, military bases, prisons, and eventually at home or in any location, through broadband Internet.
Task Progress:	During 2006, we scripted the depression treatment program, and produced the majority of it. Approximately 100 video clips and 200 audio clips are included in the program, embedded in a highly complex, branching multimedia program. We also conducted on-camera interviews with persons who have had depression, including persons who have completed Problem-Solving Treatment (PST). Our interviewees included two military pilots and an elite marine; all of whom share their observations about depression and treatment. Additionally, we interviewed one veteran long-duration ISS astronaut. This individual has not had depression, but did share his recommendations on how to maintain good psychological health on long-duration missions. Work has also progressed on setting up the clinical trial, scheduled for the next two project years.
Bibliography Type:	Description: (Last Updated: 02/11/2021)
Articles in Peer-reviewed Journals	Carter JA, Buckey JC, Greenhalgh L, Holland AW, Hegel MT. "An interactive media program for managing psychosocial problems on long-duration spaceflights." Aviat Space Environ Med. 2005 June;76(6Suppl):B213-23. PMID: 15943215 , Jun-2005